

```

mysql> /* Triggers
/*> A trigger is a special type of stored program that automatically
executes(fires) in response to the specific event.
/*> INSERT, UPDATE, DELETE occurring on a specified table. Unlike a stored
procedure. a trigger never called directly.
/*>
/*> general syntax:
/*> delimiter //
/*> create trigger trigger_name
/*> {before|after}{insert|update|delete}
/*> on table_name
/*> for each row
/*> begin
/*> --trigger logic
/*> --use new column name for inserted/updated values
/*> --use old column name for delete/previous values
/*> end //
/*> delimiter ;
/*> */
mysql> CREATE TABLE Employees (
->     emp_id INT PRIMARY KEY,
->     emp_name VARCHAR(50),
->     department VARCHAR(30),
->     salary DECIMAL(10,2),
->     experience INT,
->     city VARCHAR(30)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Employees VALUES
-> (101,'Rahul','IT',65000,5,'Hyderabad'),
-> (102,'Priya','HR',45000,3,'Chennai'),
-> (103,'Amit','Finance',70000,6,'Bangalore'),
-> (104,'Sneha','IT',80000,8,'Hyderabad'),
-> (105,'Kiran','Sales',40000,2,'Pune');
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> CREATE TABLE Employee_Audit(
->     audit_id INT AUTO_INCREMENT PRIMARY KEY,
->     emp_id INT,
->     action_type VARCHAR(20),
->     old_salary DECIMAL(10,2),
->     new_salary DECIMAL(10,2),
->     action_time DATETIME
-> );
Query OK, 0 rows affected (0.05 sec)

mysql> select * from employees;
+-----+-----+-----+-----+-----+-----+

```

emp_id	emp_name	department	salary	experience	city
101	Rahul	IT	65000.00	5	Hyderabad
102	Priya	HR	45000.00	3	Chennai
103	Amit	Finance	70000.00	6	Bangalore
104	Sneha	IT	80000.00	8	Hyderabad
105	Kiran	Sales	40000.00	2	Pune

5 rows in set (0.04 sec)

```
mysql> select * from employee_audit;
Empty set (0.00 sec)
```

```
mysql> -- before insert
mysql> delimiter //
mysql> create trigger trg_before_insert_salary
-> before insert
-> on employees
-> for each row
-> begin
-> if new.salary < 20000 then signal sqlstate'45000'
-> set message_text = 'salary cannot be less than 20000';
-> end if;
-> end //
```

Query OK, 0 rows affected (0.05 sec)

```
mysql> delimiter ;
mysql> insert into employees values(106,'Malli','IT',15000,2,'Delhi');
ERROR 1644 (45000): salary cannot be less than 20000
mysql> insert into employees values(106,'Malli','IT',25000,2,'Delhi');
Query OK, 1 row affected (0.05 sec)
```

```
mysql> -- after insert
mysql> delimiter //
mysql> create trigger trg_after_insert
-> after insert
-> on employees
-> for each row
-> begin
-> insert into employee_audit(emp_id,action_type,new_salary,action_time)
-> values(new.emp_id,'insert',new.salary,now());
-> end //
```

Query OK, 0 rows affected (0.06 sec)

```
mysql> delimiter ;
mysql> insert into employees values(107,'Kavya','HR',55000,4,'Chennai');
Query OK, 1 row affected (0.05 sec)
```

```
mysql> select * from employee_audit;
```

+-----+-----+-----+-----+-----+-----+

audit_id	emp_id	action_type	old_salary	new_salary	action_time
1	107	insert	NULL	55000.00	2026-09-24 14:43:42

1 row in set (0.00 sec)

```
mysql> -- before update
mysql> delimiter //
mysql> create trigger trg_before_update_salary
-> before update
-> on employees
-> for each row
-> begin
-> if new.salary < old.salary then signal sqlstate'45000'
-> set message_text = 'Salary cannot be lower';
-> end if;
-> end //
```

Query OK, 0 rows affected (0.06 sec)

```
mysql> delimiter ;
mysql> update employees set salary = 50000 where emp_id = 101;
ERROR 1644 (45000): Salary cannot be lower
mysql> -- after update
mysql> delimiter //
mysql> create trigger trg_after_update
-> after update on employees for each row
-> begin
-> insert into
employee_audit(emp_id,action_type,old_salary,new_salary,action_time)
-> values(new.emp_id,'update',old.salary,new.salary,now());
-> end //
```

Query OK, 0 rows affected (0.08 sec)

```
mysql> delimiter ;
mysql> update employees set salary = 70000 where emp_id = 101;
Query OK, 1 row affected (0.05 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> select * from employee_audit;
```

audit_id	emp_id	action_type	old_salary	new_salary	action_time
1	107	insert	NULL	55000.00	2026-09-24 14:43:42
2	101	update	65000.00	70000.00	2026-09-24 15:02:52

2 rows in set (0.00 sec)

```
mysql> -- before delete
mysql> delimiter //
mysql> create trigger trg_before_delete_it
```

```

-> before delete on employees for each row
-> begin
-> if old.department = 'IT' then signal sqlstate'45000'
-> set message_text = 'cannot delete IT employees';
-> end if;
-> end //

```

Query OK, 0 rows affected (0.05 sec)

```

mysql> delimiter ;
mysql> delete from employees where emp_id = 101;
ERROR 1644 (45000): cannot delete IT employees
mysql> -- after delete
mysql> delimiter //
mysql> create trigger trg_after_delete
-> after delete on employees for each row
-> begin
-> insert into employee_audit(emp_id, action_type, old_salary, action_time)
-> values(old.emp_id,'delete',old.salary,now());
-> end //

```

Query OK, 0 rows affected (0.07 sec)

```

mysql> delimiter ;
mysql> delete from employees where emp_id = 102;
Query OK, 1 row affected (0.05 sec)

```

```

mysql> select * from employee_audit;

```

audit_id	emp_id	action_type	old_salary	new_salary	action_time
1	107	insert	NULL	55000.00	2026-09-24 14:43:42
2	101	update	65000.00	70000.00	2026-09-24 15:02:52
3	102	delete	45000.00	NULL	2026-09-24 15:11:50

3 rows in set (0.00 sec)

```

mysql> notee

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